

Hardware: The CPU & Storage

Chapter 4



4

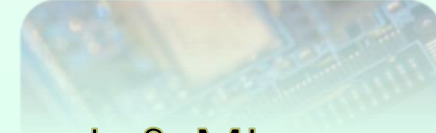
Chapter Topics

UNIT 4A: Processing: The System Unit, Microprocessors, & Main Memory

- 4.1 Microchips, Miniaturization, & Mobility
- 4.2 Representing Data Electronically
- 4.3 Inside the System Unit: Power Supply, Motherboard, & Microprocessors
- 4.4 The Central Processing Unit & the Machine Cycle
- 4.5 Memory
- 4.6 Expansion Cards, Bus Lines, & Ports

UNIT 4B: Secondary Storage

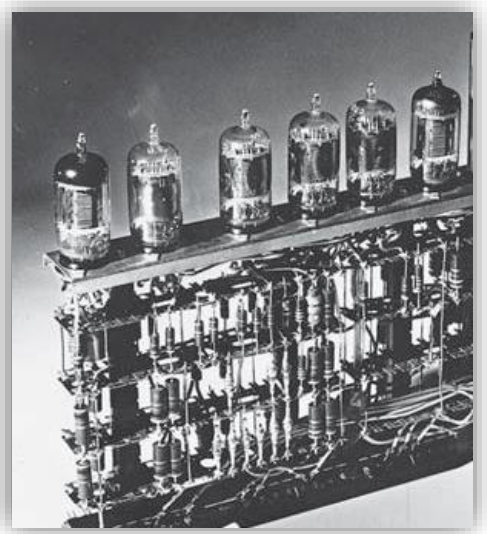
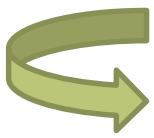
- 4.7 Secondary Storage
- 4.8 Future Developments in Processing & Storage





UNIT 4A: Processing: The System Unit, Microprocessors, & Main Memory

- Electronic circuitry has remained basically the same over recent years.
- A **circuit** is a closed path followed or capable of being followed by an electric current.
- Vacuum tubes used wire circuits inside them to facilitate the flow of electrons.
- Transistors have replaced vacuum tubes.





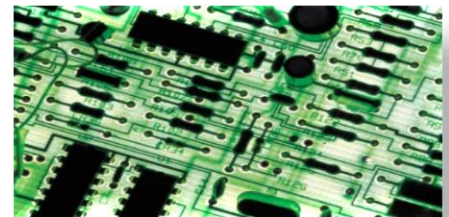
4.1 Microchips, Miniaturization, & Mobility



The Since the early 1970s, microchips have gotten smaller and smaller yet more and more powerful and faster.



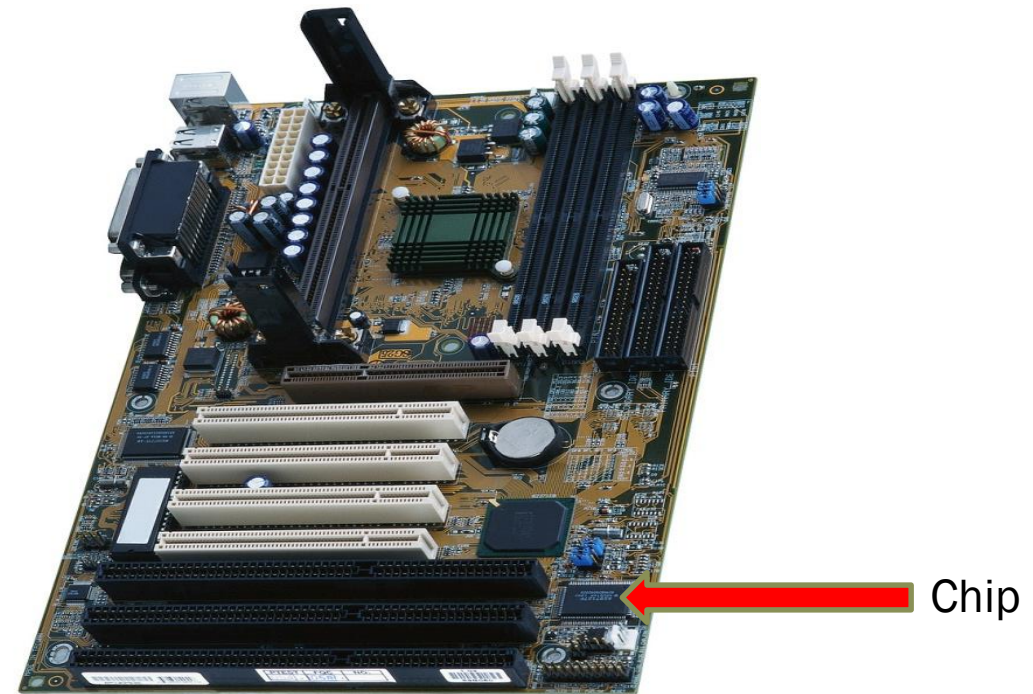
- A **transistor** is a tiny electronic switch that can be turned “on” or “off” millions of times per second.
- Transistors form part of an **integrated circuit**: all the parts of an electronic circuit embedded on a single silicon chip.
- Integrated circuits are **solid state** (no moving parts).



Si Silicon

Atomic Number: 14
Atomic Mass: 28.09

- **Silicon:** A semiconductor made of clay and sand.
- **Semiconductor:** A material whose electrical properties are intermediate between a good conductor and a nonconductor of electricity.
 - Perfect underlayer for highly conductive, complex circuits.
 - Microchips (Microprocessors) are made from semiconductors.
- **Chip:** A tiny piece of silicon that contains millions of microminiature integrated electronic circuits.





- **Miniaturization**

- **Microchips**

- Store and process data in electronic devices

- **Microprocessors**

- The miniaturized circuitry of an entire computer processor (“brain”) on a single chip
 - Contains the central processing unit (**CPU**), which processes data into information

- The development of microchips and processors has enabled the development of small, mobile electronic devices.



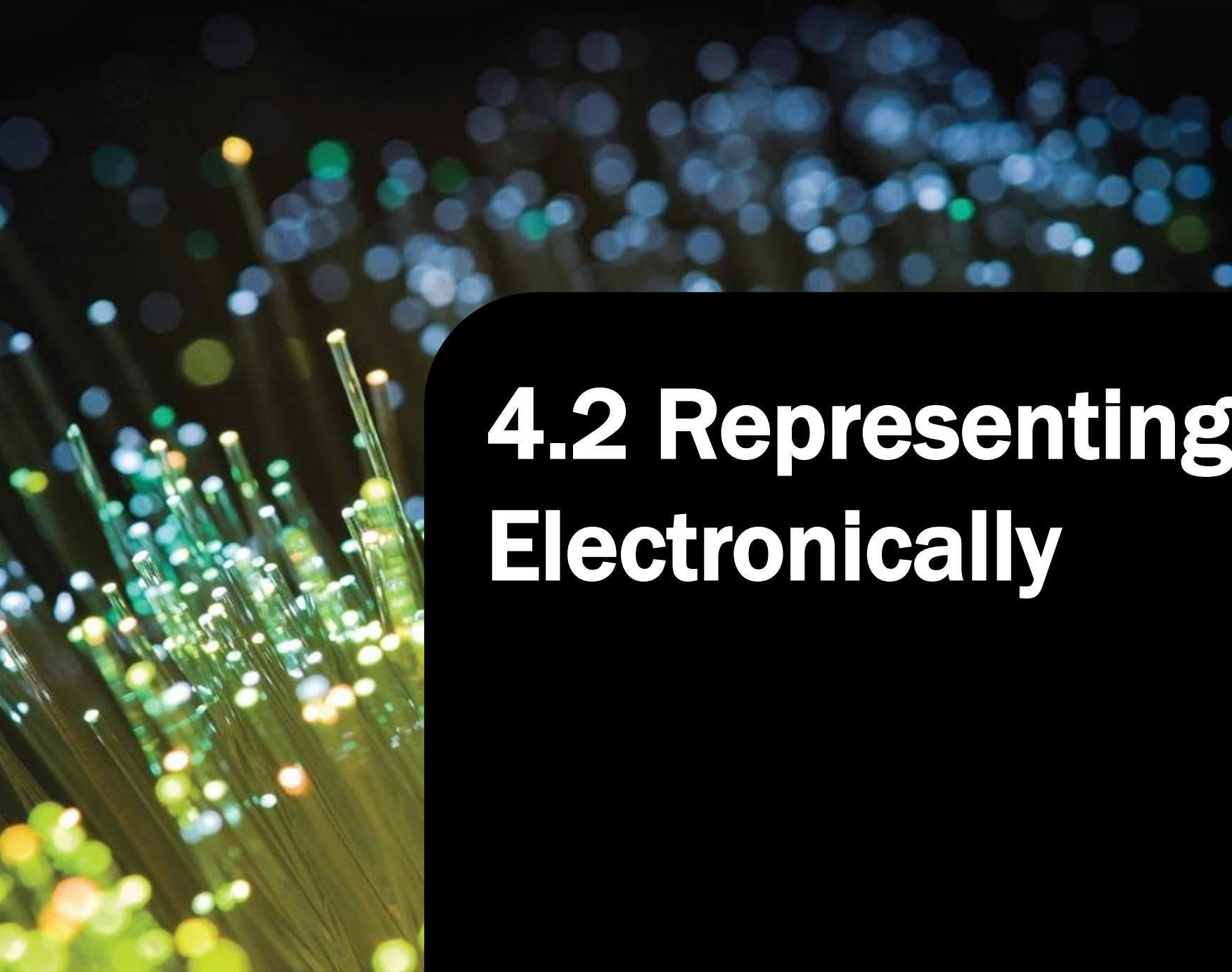


• System Unit

- The case that contains the computer's electronic components used to process data.

- PCs: Tower or desktop; monitor is separate.
- Laptops: Monitor is attached to the system unit, like a clamshell.
- Tablets: Usually includes a touch-screen interface.
- Smartphones: Handheld system units.



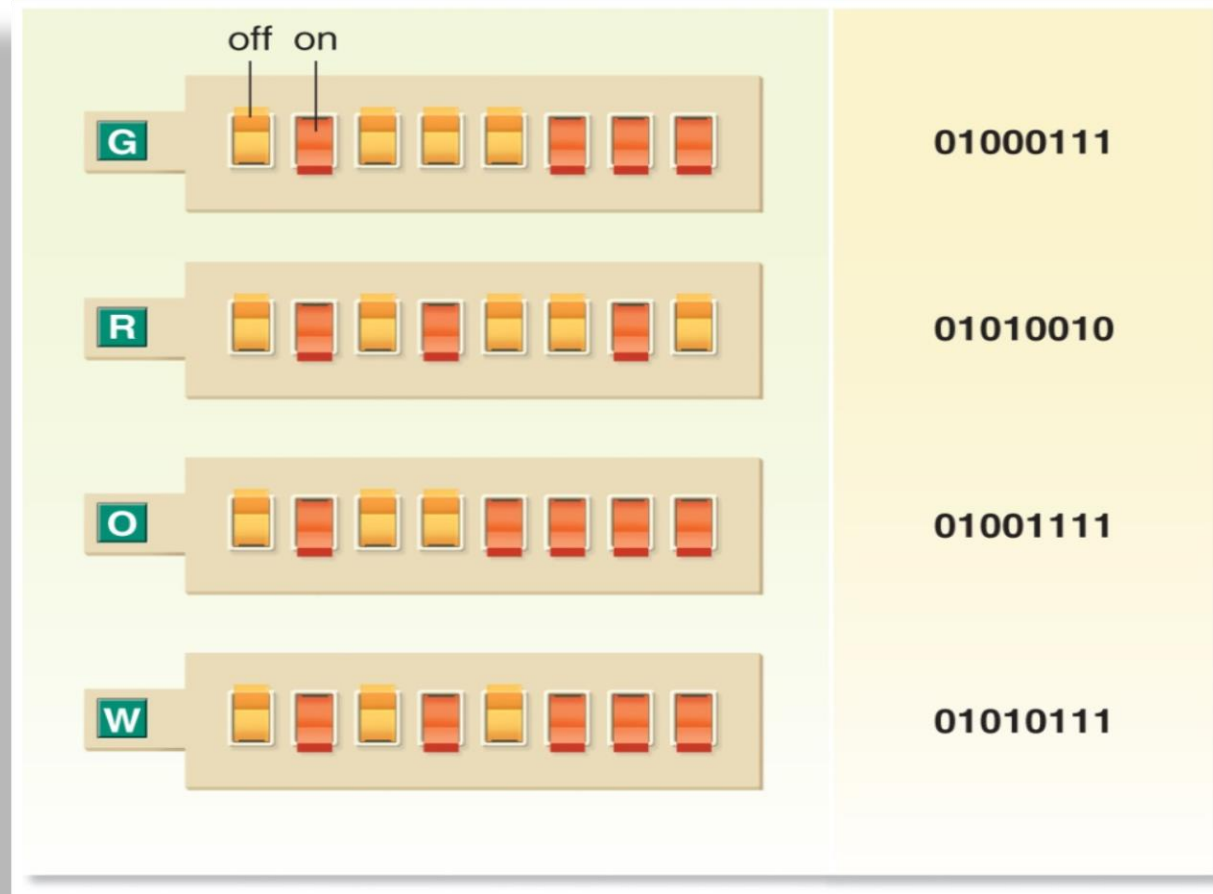


4.2 Representing Data Electronically



Data is represented in a computer by binary code.

Binary System: the basic data-representation method for computers uses just two numbers: 0 and 1, representing the off/on states of electricity or light pulses.

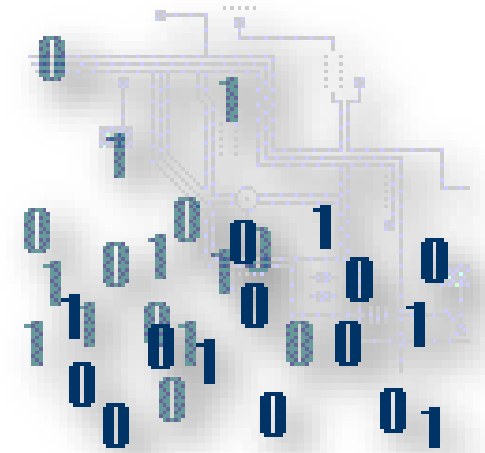




Measuring Capacity

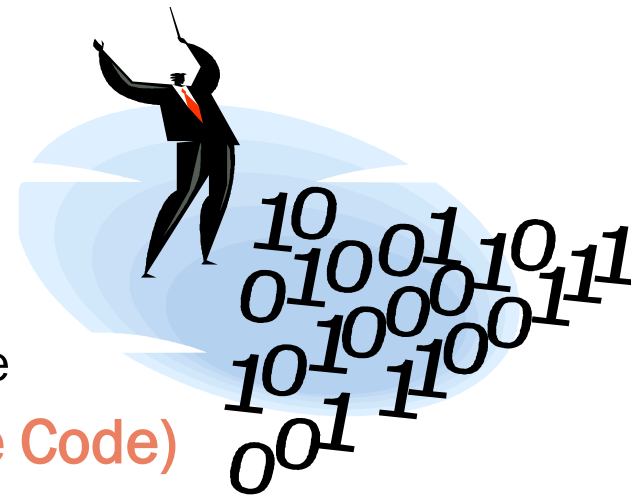
All data and program instructions in the computer are represented as binary.

- **Bit:** each 0 or 1 is a bit
- **Byte:** a group of 8 bits = 1 character, digit, or other value
- **Kilobyte (KB):** 1,000 (1,024) bytes
- **Megabyte (MB):** 1 million (1,048,576) bytes
- **Gigabyte (GB):** 1 billion (1,073,741,824) bytes
- **Terabyte (TB):** 1 trillion (1,009,511,627,776) bytes
- **Petabyte (PB):** 1 quadrillion bytes
- **Exabyte (EB):** 1 quintillion bytes





- Binary coding schemes assign a unique binary code to each character.
 - **ASCII (American Standard Code for Information Interchange)**
 - Requires 7 or 8 bits per character, depending on the version
 - 8-bit Extended ASCII provides 256 characters
 - Commonly used for microcomputers
 - **Unicode**
 - Requires 16 bits per character
 - Handles 65,536 characters—used for Chinese and Japanese
 - **EBCDIC (Extended Binary Coded Decimal Interchange Code)**
 - Requires 8 bits per character
 - Used for IBM mainframes





Character	ASCII-8	EBCDIC	Character	ASCII-8	EBCDIC
A	0100 0001	1100 0001	N	0100 1110	1101 0101
B	0100 0010	1100 0010	O	0100 1111	1101 0110
C	0100 0011	1100 0011	P	0101 0000	1101 0111
D	0100 0100	1100 0100	Q	0101 0001	1101 1000
E	0100 0101	1100 0101	R	0101 0010	1101 1001
F	0100 0110	1100 0110	S	0101 0011	1110 0010
G	0100 0111	1100 0111	T	0101 0100	1110 0011
H	0100 1000	1100 1000	U	0101 0101	1110 0100
I	0100 1001	1100 1001	V	0101 0110	1110 0101
J	0100 1010	1101 0001	W	0101 0111	1110 0110
K	0100 1011	1101 0010	X	0101 1000	1110 0111
L	0100 1100	1101 0011	Y	0101 1001	1110 1000
M	0100 1101	1101 0100	Z	0101 1010	1110 1001
0	0011 0000	1111 0000	5	0011 0101	1111 0101
1	0011 0001	1111 0001	6	0011 0110	1111 0110
2	0011 0010	1111 0010	7	0011 0111	1111 0111
3	0011 0011	1111 0011	8	0011 1000	1111 1000
4	0011 0100	1111 0100	9	0011 1001	1111 1001
!	0010 0001	0101 1010	;	0011 1011	0101 1110



- **Machine Language:** The computer’s “native language”
 - A binary-type programming language (0s and 1s) built into the CPU that is run directly by the computer.
 - Each CPU type has its own machine language; thus each computer’s machine language is brand-dependent.
 - **Language Translators:** The computer’s system software converts higher-level language instructions and data into machine language so that the processor can “understand” what to do.



4.3 Inside the System Unit

Power Supply, Motherboard, & Microprocessors

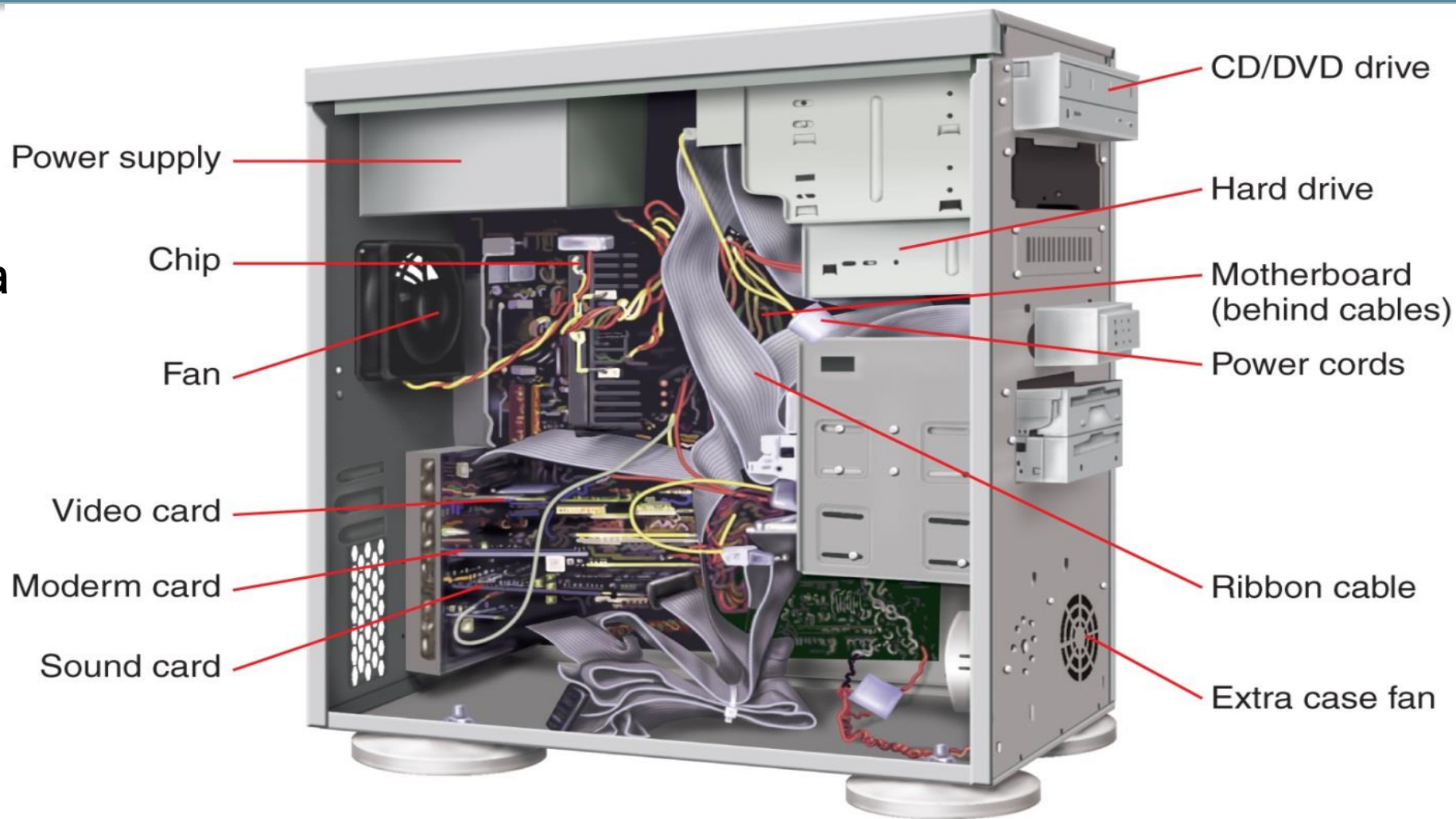


Terms

Definitions

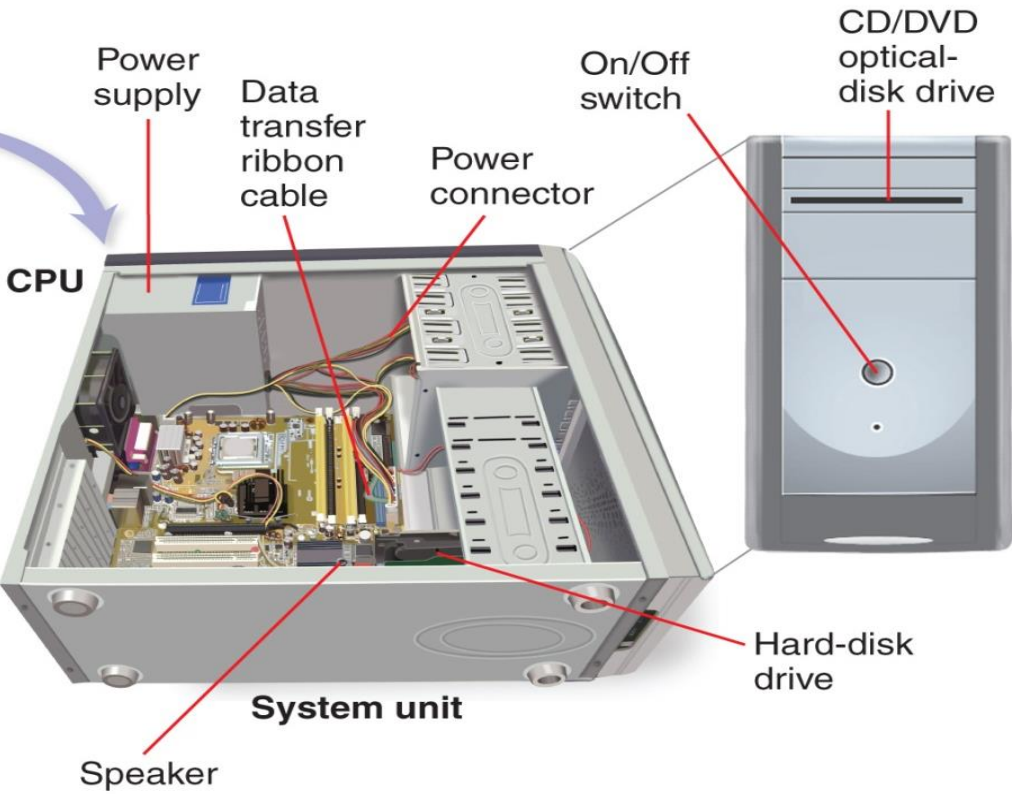
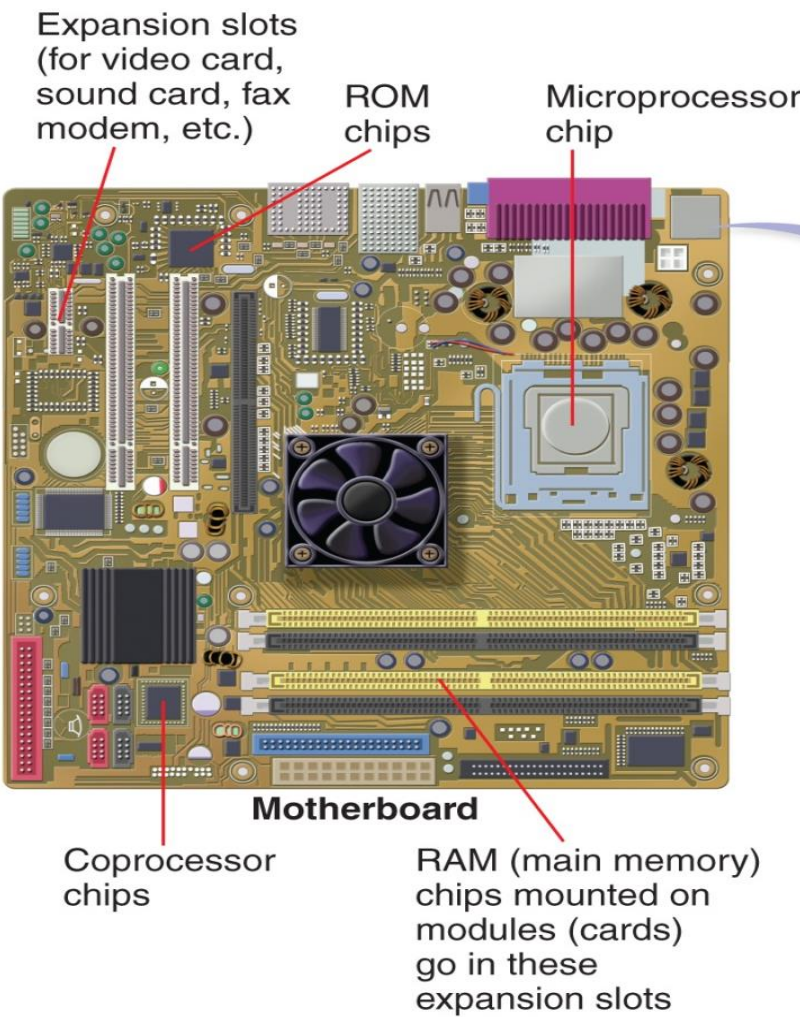
- Bay** → Opening in the computer cabinet used for the installation of electrical equipment.
- Power Supply** → This converts AC to DC to run the computer.
- Surge Protector** → Protects the computer from being damaged by power spikes. Plug your computer into one.
- UPS** → Uninterruptible Power Supply. Battery-operated device that provides power for a limited time when there is a blackout.
- Motherboard** → Main system board of the computer (also *systemboard*).
- Microprocessor** → Miniaturized circuitry of a computer processor.
- Chipset** → Groups of interconnected chips on the motherboard that control information flow between the microprocessor and other system components connected to the motherboard.

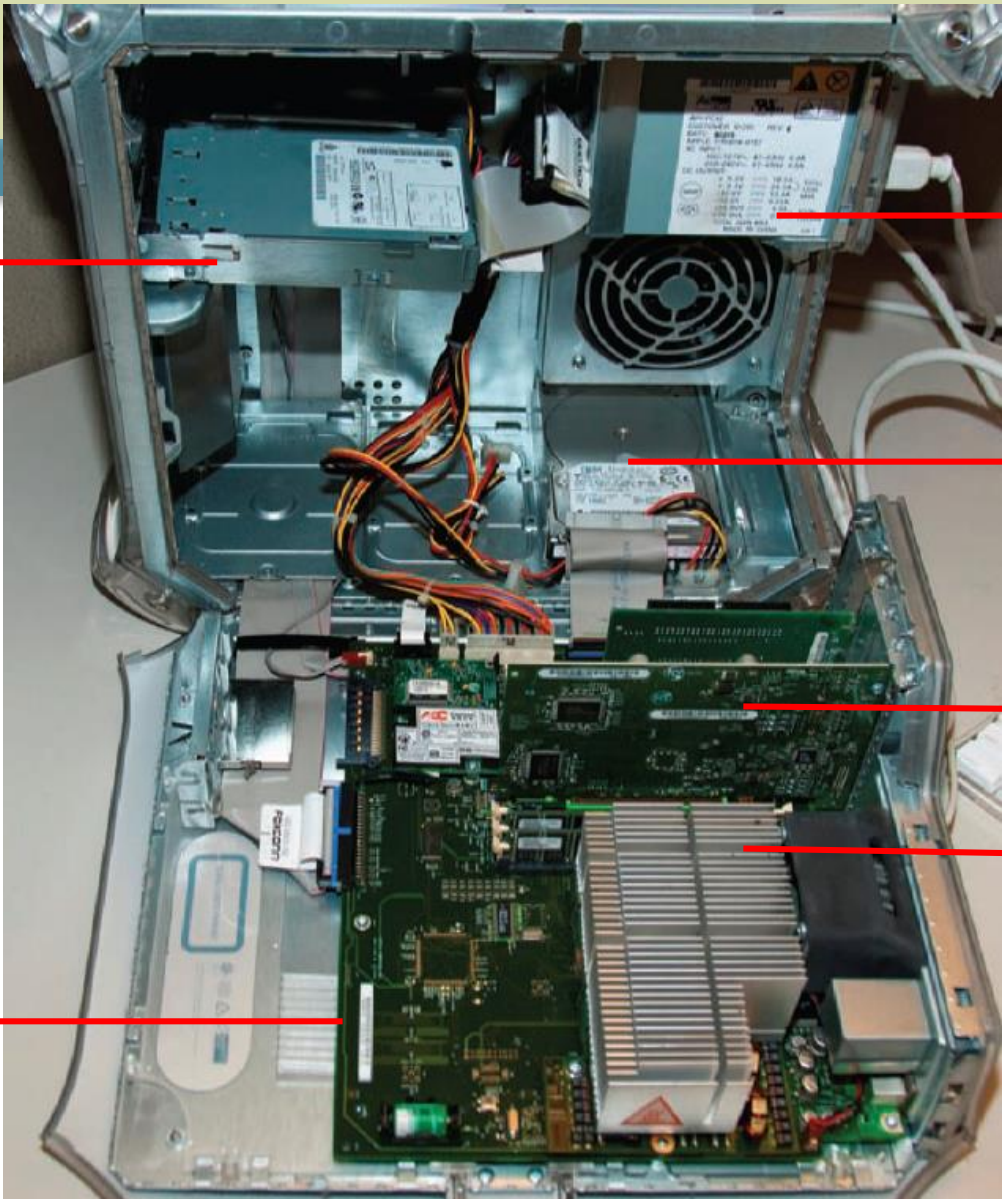
One View of a Motherboard





Another View of a Motherboard





CD drive

Power supply

A Mac
motherboard

Hard drive

Expansion
cards

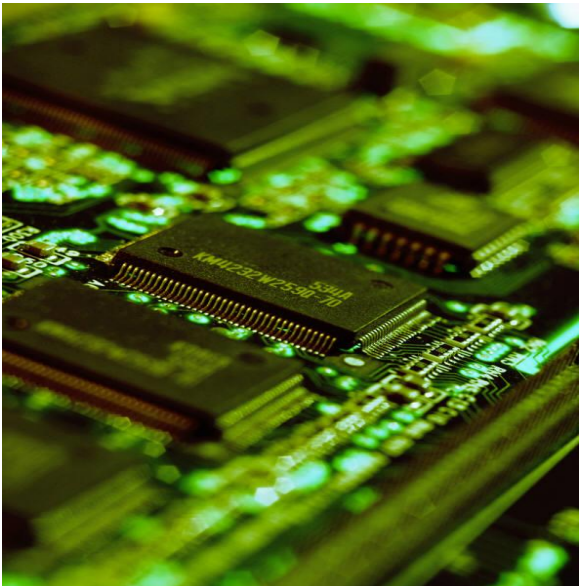
RAM slots

Motherboard





- Traditional microcomputer microprocessors are **Intel** and **AMD**.
- **Multicore processors** (2, 4, 6, 8) have more than one processor “core” on a single silicon chip, which allows computers to run faster.
 - Special processors are made for portable devices.
 - A **graphics processing unit (GPU)** is a specialized processor used to manipulate three-dimensional (3-D) computer graphics.





Processing Speeds

- Every microprocessor contains a **system clock**, which controls how fast all the operations within a computer take place (the chip's processing speed).
- Older CPU processing speeds are in megahertz.
 - 1 **MHz** = 1 million cycles per second
- Current CPU processing speeds are in gigahertz.
 - 1 **GHz** = 1 billion cycles per second
- The faster a CPU runs, the more power it consumes, and the more heat it generates.

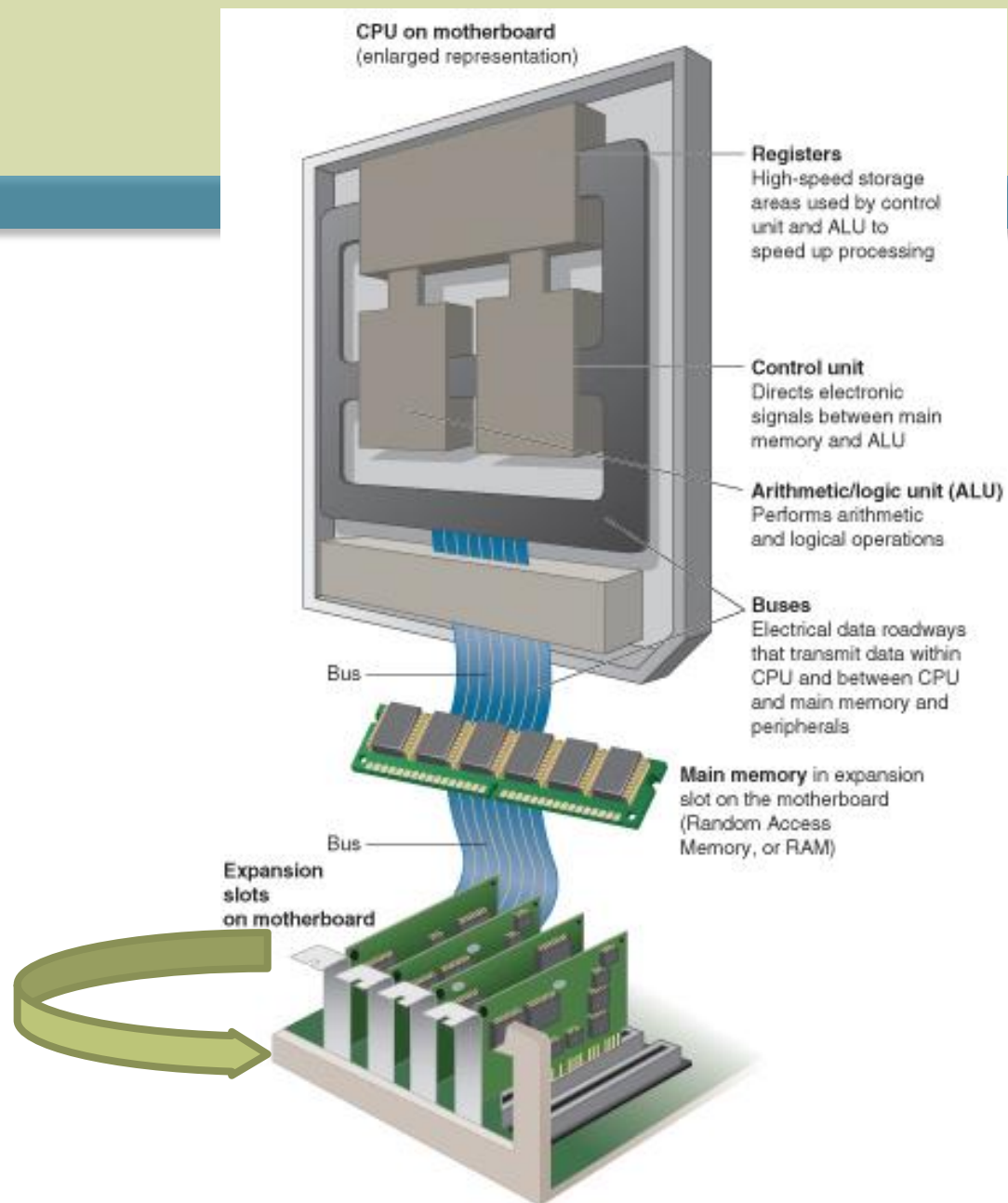




4.4 The Central Processing Unit & the Machine Cycle



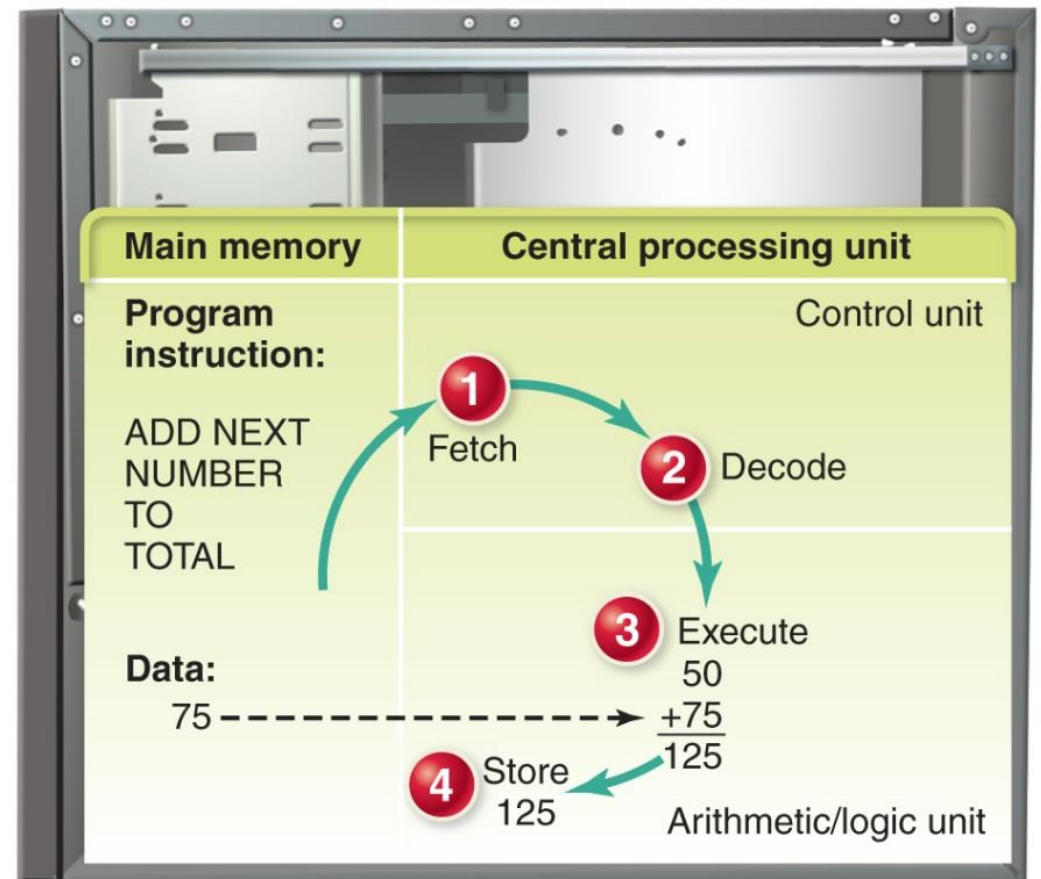
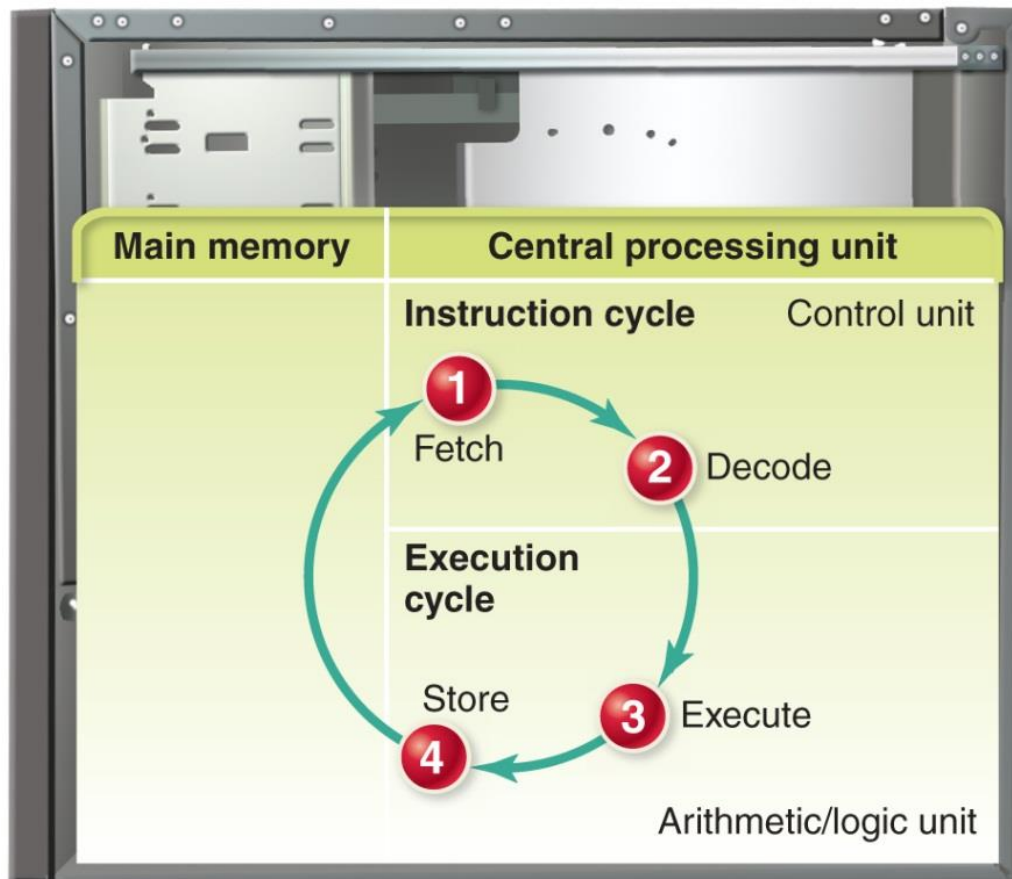
- The **CPU**, for **central processing unit**, is the “brain” of the computer; it follows the instructions of the software (program) to manipulate data into information.
- The CPU consists of two parts—(1) the **control unit** and (2) the **arithmetic/logic unit (ALU)**, both of which contain **registers**, or high-speed storage areas.
- All are linked by a kind of electronic “roadway” called a **bus**.





- The control unit deciphers each instruction stored in the CPU and then carries out the instruction. It directs the movement of electronic signals between main memory and the arithmetic/logic unit. It also directs these electronic signals between main memory and the input and output devices.
- For every instruction, the control unit carries out four basic operations, known as the **machine cycle**. In the machine cycle, the CPU (1) fetches an instruction, (2) decodes the instruction, (3) executes the instruction, and (4) stores the result (*see next slide*)..

Each time the central processing unit executes an instruction, it takes a series of steps. The complete series of steps is called a **machine cycle**.





- The **arithmetic/logic unit (ALU)** performs arithmetic operations and logical operations and controls the speed of those operations.
- *Arithmetic operations* are the fundamental math operations: addition, subtraction, multiplication, and division.
- *Logical operations* are comparisons--the ALU compares two pieces of data to see whether one is equal to (=), greater than (>), greater than or equal to (>=), less than (<), less than or equal to (<=), or not equal to (≠) the other.



- The control unit and the ALU also use **registers**, special CPU areas that enhance the computer's performance.
- Registers are high-speed storage areas that temporarily store data during processing. They may store a program instruction while it is being decoded, store data while it is being processed by the ALU, or store the results of a calculation.
- All data must be represented in a register before it can be processed.
- The number of registers that a CPU has and the size of each (number of bits) help determine the power and speed of a CPU.



- **Buses** are electrical data “roadways” through which bits are transmitted within the CPU and between the CPU and other components of the motherboard.
- In most computers, the bus width is the same as the computer’s **word size**, the number of bits that the processor can process at any one time. The more bits in a word, usually the faster the computer. A 32-bit-word computer will transfer data within each microprocessor chip in 32-bit chunks. A 64-bit-word computer is faster, transferring data in 64-bit chunks at a time. (Most, but not all, 32-bit software will run on a 64-bit system, but 64-bit software will not run on a 32-bit system.)

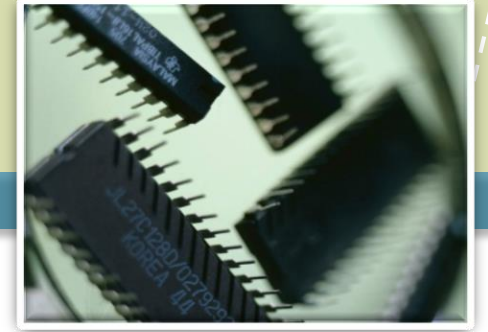


4.5 Memory



Memory

- Two type of storage: **primary** and **secondary**.
 - Primary storage = “memory,” “main memory,” “RAM”; this type of memory is temporary and volatile.
 - Secondary storage = hard disks and flash memory units; this type of memory is relatively permanent and nonvolatile.



Memory Chips

RAM

Random Access Memory chips are **volatile** and hold:

- a. Software instructions
- b. Data before & after the CPU processes it

ROM

Read Only Memory

- a. Cannot be written on or erased without special equipment
- b. Are loaded at factory with fixed (permanent) start-up instructions (BIOS), that tell the computer how to load the operating system

CMOS

Complementary Metal Oxide Semiconductor

- a. Powered by a battery
- b. Contains time, date, calendar, boot password

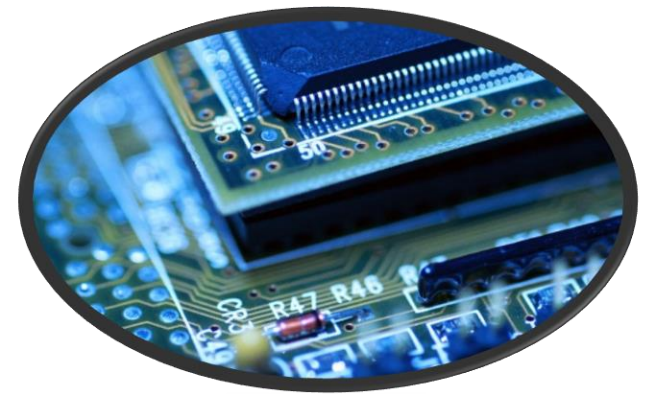
Flash

Nonvolatile memory that can be erased and reprogrammed more than once

- a. Doesn't require a battery
- b. Used in newer PCs for BIOS instructions

Speeding up Processing: **Cache**

- The CPU works much faster than RAM, so it often must wait for information
- **Cache** temporarily stores instructions and data that the processor uses frequently to speed up processing
 - **Level 1** cache is part of the microprocessor
 - Holds 8 to 128 KB
 - Faster than Level 2 cache
 - **Level 2** cache is external cache
 - Holds 64 kb to 16 MB
 - **Level 3** cache is on the motherboard
 - Comes on very high-end computers
 - Holds 2 to 8 MB



- **Virtual Memory**—also used to speed up processing.
 - This type of memory is unused hard disk or optical (CD) space that the processor uses to extend the capacity of RAM.
 - The processor goes first to L1 cache, then L2 cache, then RAM, then virtual memory.
 - Each type of memory is slower than its predecessor.





4.6 Expansion Cards, Bus Lines, & Ports

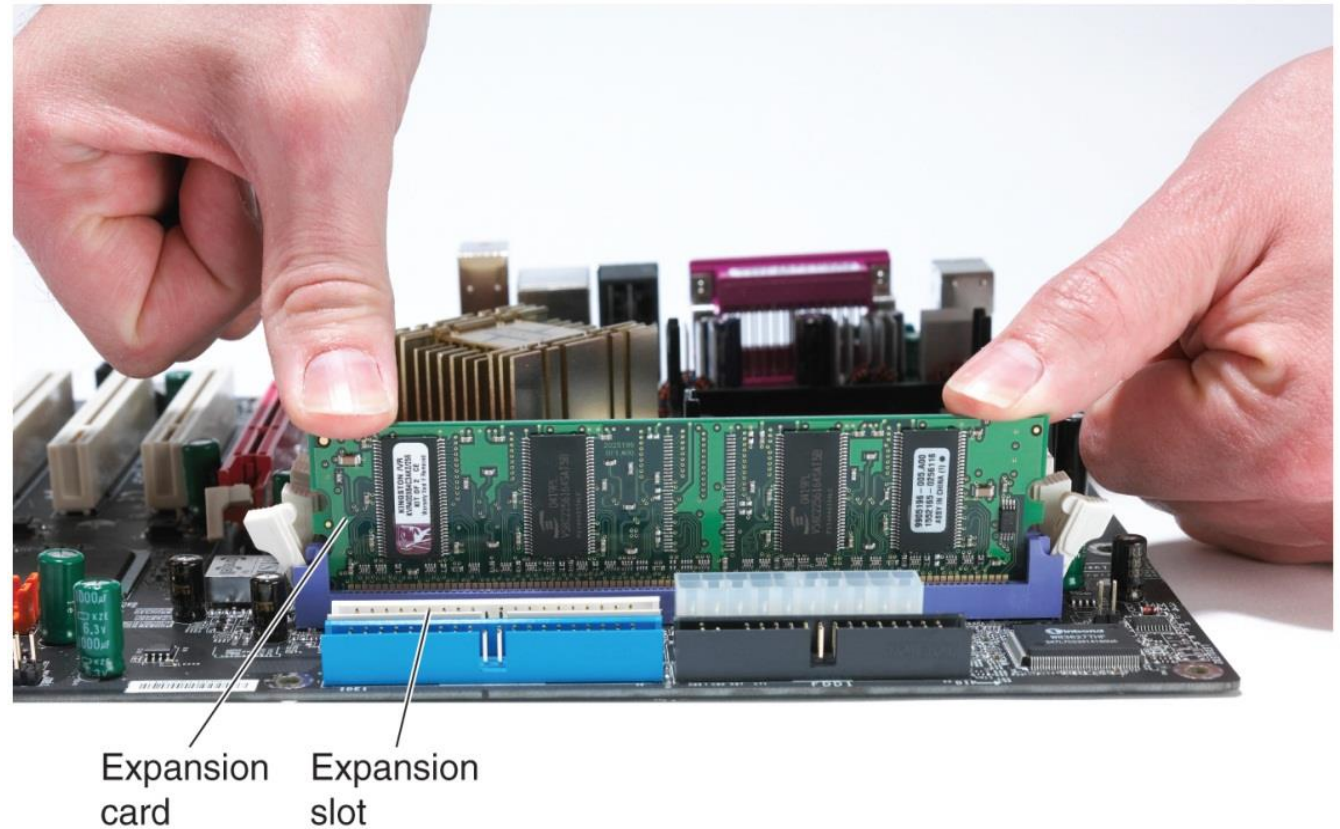


Expansion cards plug into slots on the motherboard that are connected by buses to ports that the user can access.

- **Expansion cards** are circuit boards that provide more memory or that control peripheral devices (for *graphics, sound, video, network interface, wireless connection*, etc.).
- **Buses** connect the expansion cards to ports.
- A **port** is a connecting socket or jack on the outside of the computer unit or device into which are plugged different kinds of cables that connect peripheral devices.



Expansion Cards: If a computer uses *closed architecture*, no expansion cards can be added; if the computer uses *open architecture*, expansion cards can be inserted in expansion slots inside the computer, connected to the motherboard.





An expansion bus is not the same as the frontside bus:

- **Frontside bus:** The bus that connects the CPU within itself and to main memory.
- **Expansion bus:** Buses that connect the CPU with expansion slots on the motherboard and thus via ports with peripheral devices.



Types of expansion buses:

- **PCI:** High-speed bus that has been widely used to connect PC graphics cards, sound cards, modems, and high-speed network cards.
- **PCI Express:** Doubles the speed of the original PCI bus . PCIe is the latest standard for expansion cards available on mainstream personal computers.
- **Accelerated Graphics:** Transmits data at twice the speed of a PCI bus and is designed to support video and 3-D graphics.
- **Universal Serial Bus (USB):** Does away with the need to install cards in expansion slots. USB devices can connect one to another outside the system unit, and then the USB bus connects to the PCI bus on the motherboard.
- **Firewire:** Resembles the USB bus but is used for more specialized purposes, such as to connect audio and video equipment to the motherboard.

A port is a socket for some kind of plug, of which there are many types.

Port Type

Description

Serial port

Used to transmit data slowly over long distances

- a. Sends data sequentially, one bit at a time
- b. Used to connect older keyboards, mouse, monitors, dial-up modems

Parallel port

For transmitting data quickly over short distances

- a. Transmits 8 bytes simultaneously
- b. Connects printers, external disks, tape backups

USB port



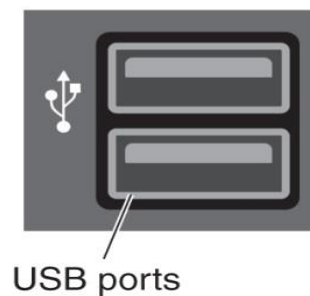
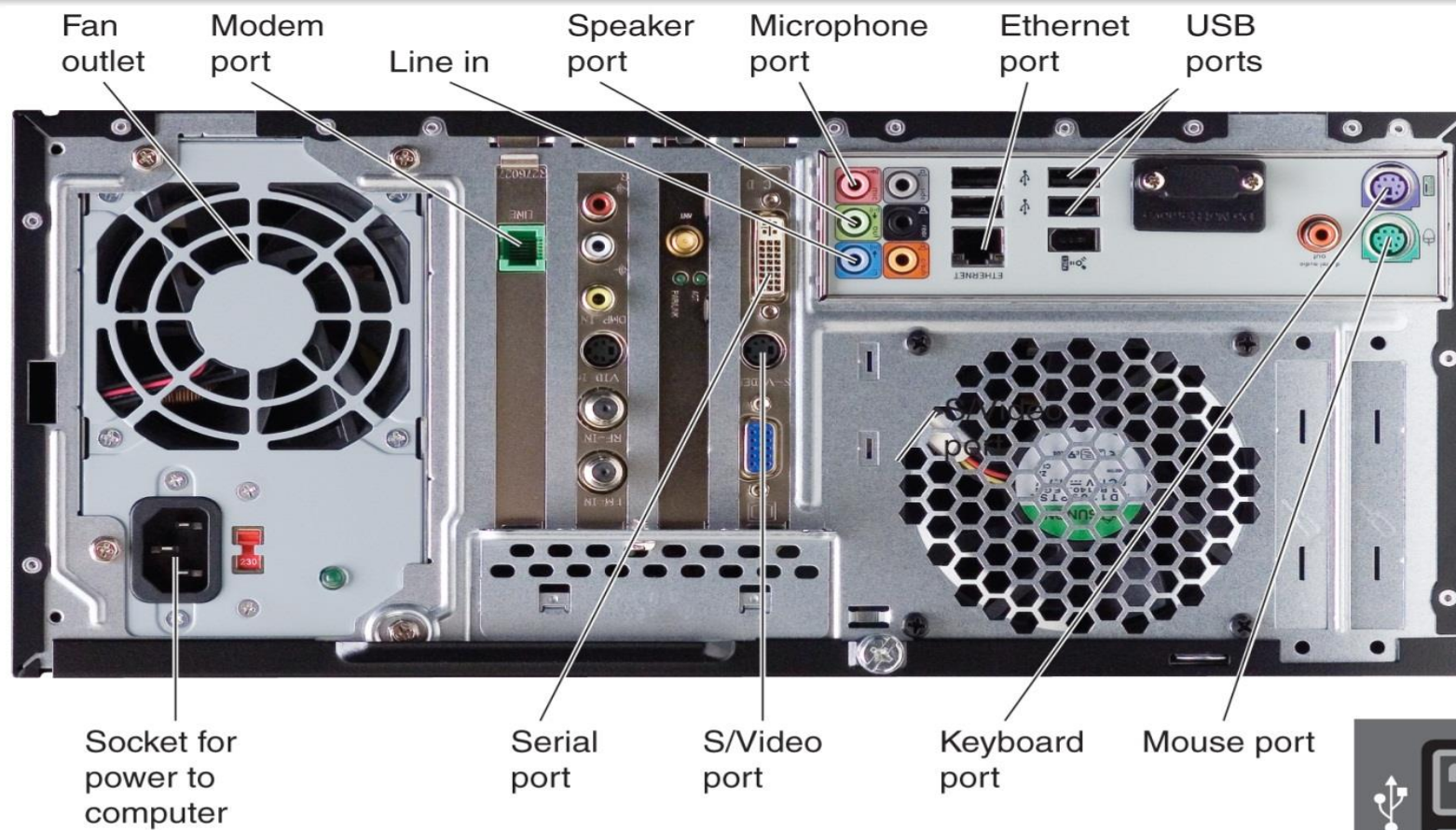
Universal Serial Bus high-speed hardware standard for interfacing peripheral devices, such as scanners and printers, to computers without a need for special expansion cards or other hardware modifications to the computer. USB is replacing many varieties of serial and parallel ports.

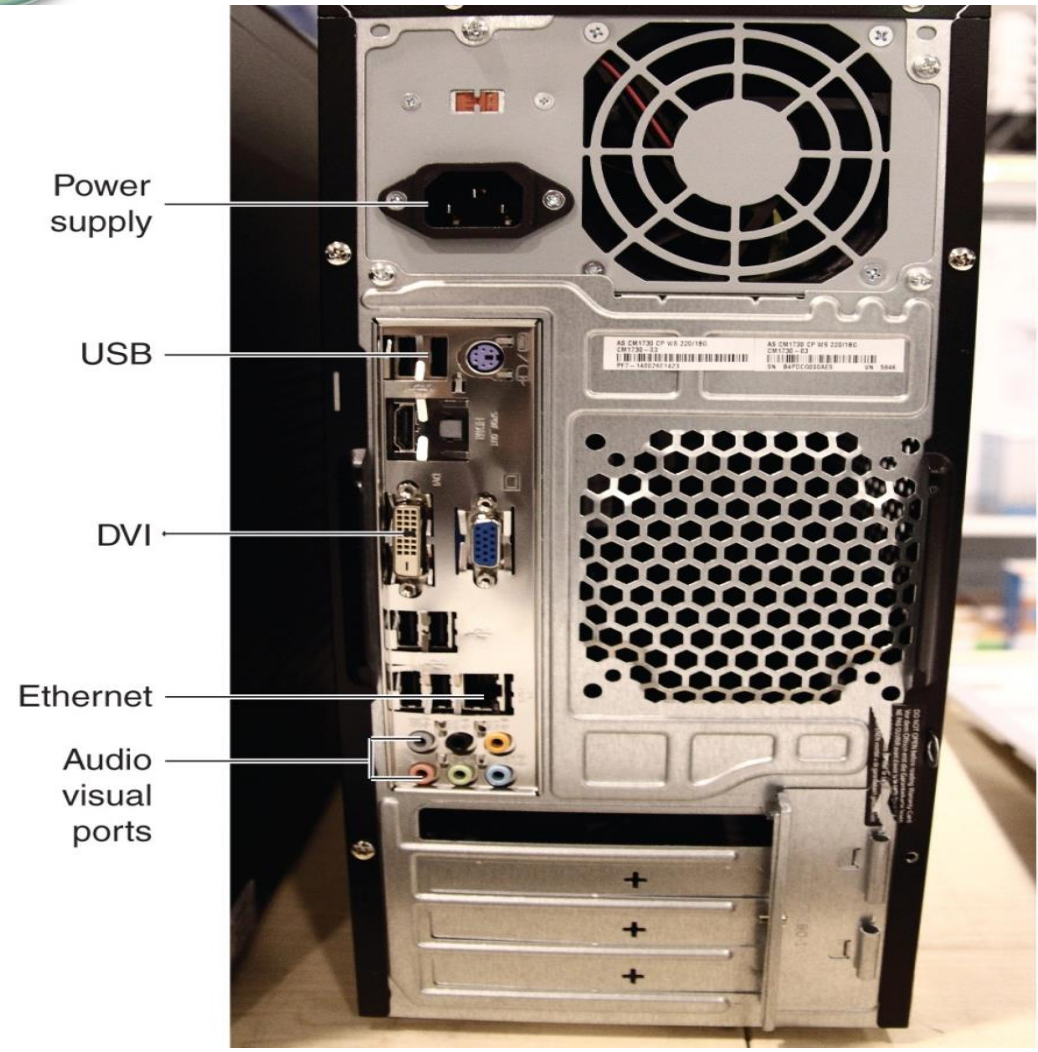


Specialized Expansion Ports

Port Type

- **FireWire** Intended for multiple devices working with lots of data and requiring fast transmission speeds, such as DVD drives, digital video cameras, and gaming consoles.
- **Ethernet** Supports a network standard for linking a wired local area network and connecting it to a DSL or a cable modem for high-speed Internet access.
- **Graphics** Connects digital monitors and multimedia digital devices, such as TVs and DVD players.
- **eSATA** External Serial Advanced Technology Attachment; allows the attachment of an eSATA hard disk, which has fast data transmission speeds.
- **Bluetooth** Connects devices that use short-range radio waves that transmit up to 30 feet.
- **IrDA** Transfers data via infrared light waves between directly aligned devices, as between a smartphone and a desktop computer.
- **HDMI** High-Definition Multimedia Interface; carries both video and audio signals and is used for connecting HDTVs, DVD players, and game consoles to computers, laptops, and other devices.
- **MIDI** Musical Instrument Digital Interface; used to connect electronic musical instruments to a sound card that converts the signals to digital instructions that can be saved or manipulated.





DVI (Digital Video Interface)

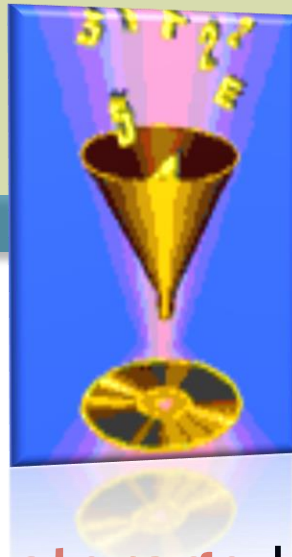
Fire Wire

USB

Head-phones



HDMI Port



UNIT 4B:

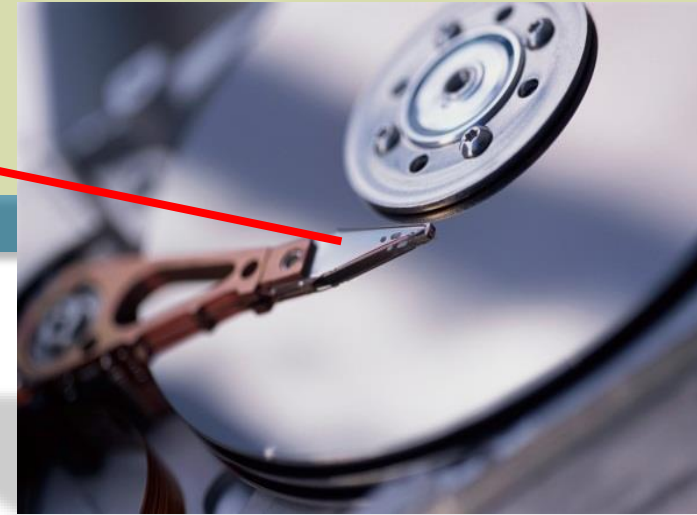
- **Secondary storage** hardware includes devices that permanently hold data and information as well as programs.
- Online, or **cloud, storage** is also available, but we still use secondary storage hardware.



4.7 Secondary Storage

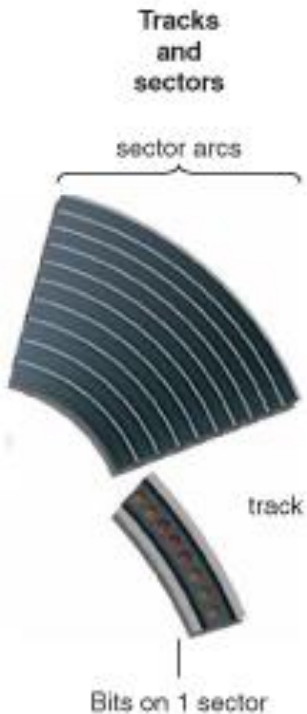


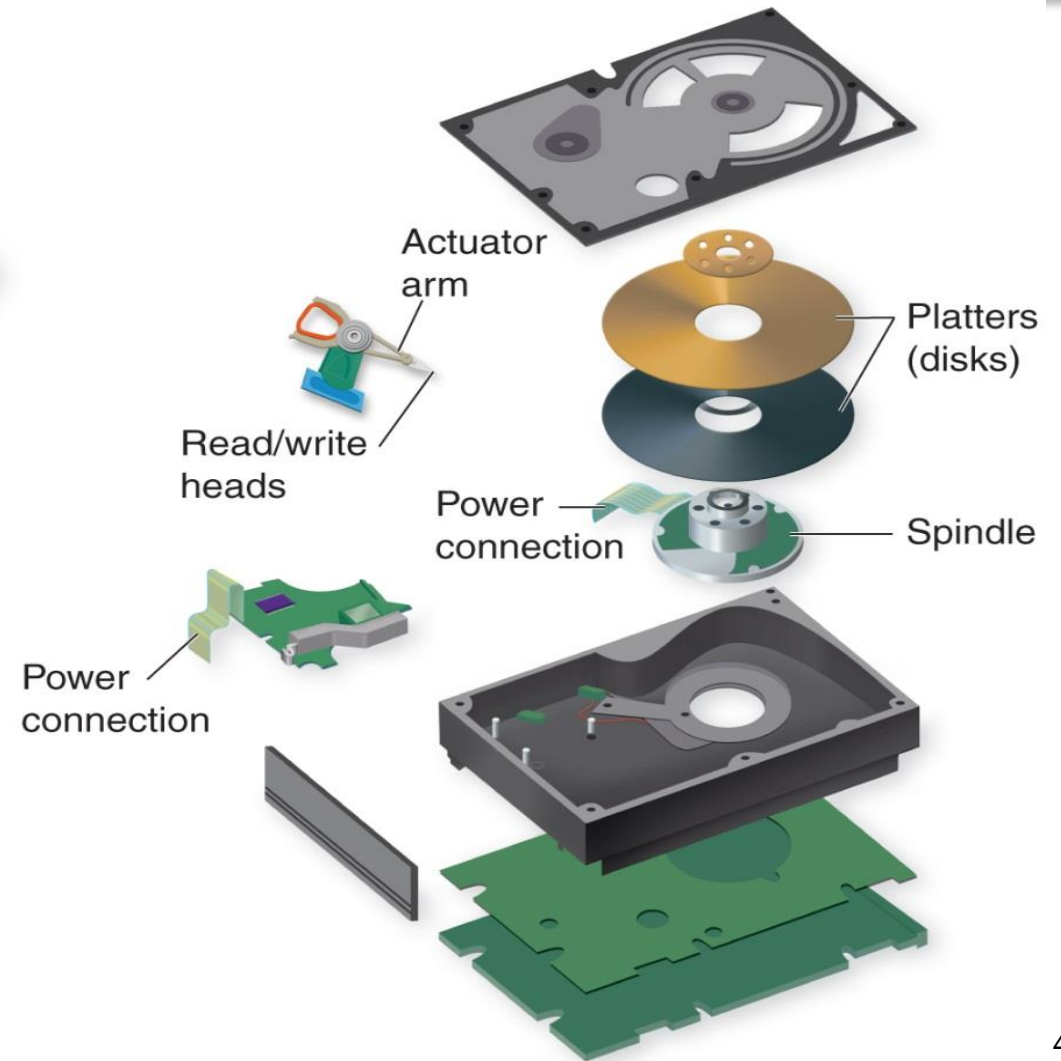
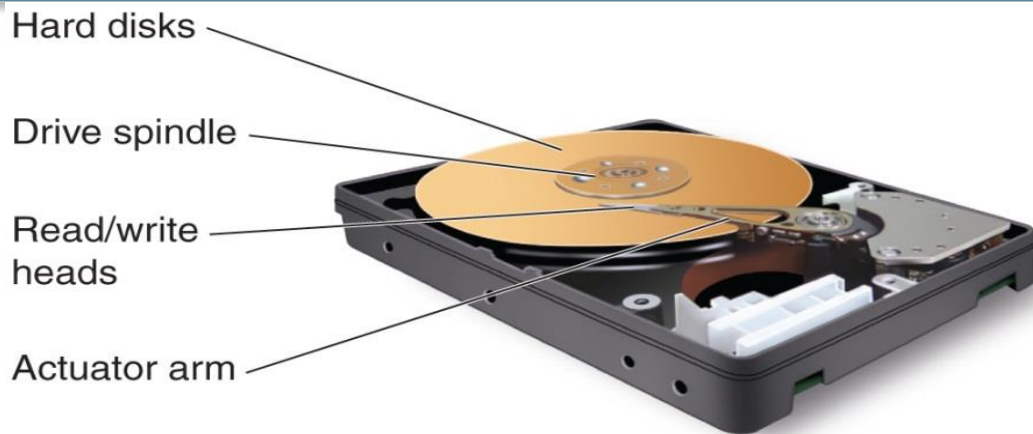
read/write
head



Hard Disks: Still the major secondary-storage for desktop/tower computers.

- Thin, rigid metal, glass, or ceramic platters covered with a substance that allows data to be held in the form of magnetized spots.
 - The more platters there are, the higher the drive capacity.
 - Store data in tracks, sectors, and clusters.
 - Formatting creates a file allocation table that maps files to clusters.
 - Drive heads ride on .000001" cushion of air, and can crash!
 - Important data should always be backed up!



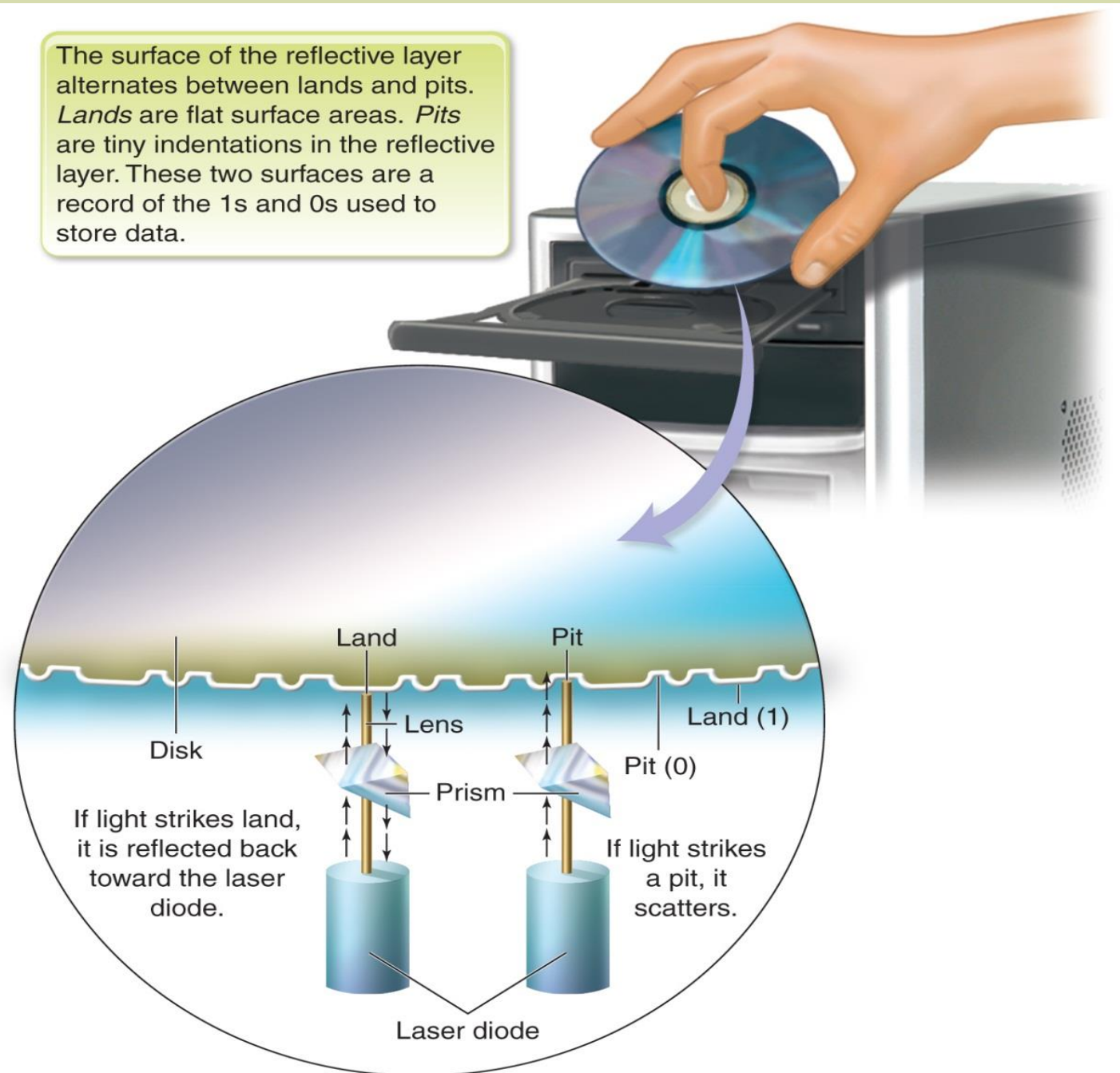




- Hard Disks (*continued*)
 - Hard Disk Types:
 - **Nonremovable hard disk** – Also known as a fixed disk; is housed in the microcomputer system unit and is used to store nearly all programs and most data files. Usually consists of several metallic or glass platters, from 1 to 5.25 inches (most commonly 3.5 inches) in diameter, stacked on a spindle, with data stored on both sides. Read/write heads, one for each side of each platter, are mounted on an access arm that moves back and forth to the right location on the platter.
 - **External hard disk** – Freestanding disk drive (portable); usually connected via USB.
 - **RAID** – redundant array of independent disks; for large computer systems.



The surface of the reflective layer alternates between lands and pits. *Lands* are flat surface areas. *Pits* are tiny indentations in the reflective layer. These two surfaces are a record of the 1s and 0s used to store data.





- **Optical Disks**

- **CDs** (compact disks) and **DVDs** (digital versatile/video disks) are optical disks.
- DVDs hold more data than CDs do.
- Data is written and read using lasers, not a disk read/write head.
 - **CD-ROM** is Compact Disk Read-Only Memory; content is prerecorded.
 - **CD-R** (compact disk-recordable) is used for recording only once.
 - **CD-RW** (compact disk-rewritable) is an erasable optical disk that can both record and erase data over and over again.

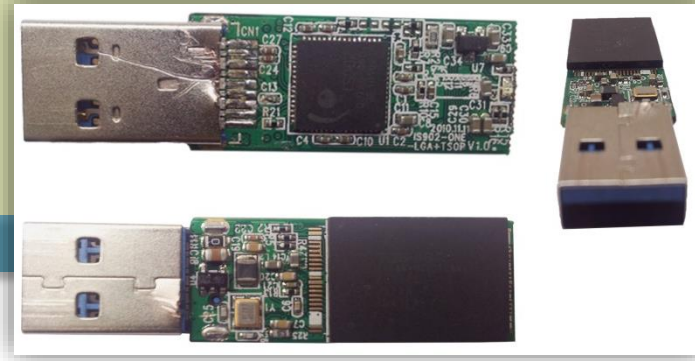


- Optical Disks (*continued*)
 - **DVD** is a CD-style disk with extremely high capacity.
 - **DVD-R** (DVD-recordable) is used for recording only once.
 - **DVD-RW, DVD-RAM, DVD+RW** are reusable DVDs.
 - **Blu-ray** is an optical-disk format used to record, rewrite, and play back high-definition (HD) video, as well as to store large amounts of data.



Flash & Solid-State Storage

- Flash memory and solid-state memory have become the most important form of mobile secondary storage.
- Disk drives (hard disks or CDs/DVDs) all involve some moving parts—and moving parts can break. By contrast, **flash memory** has no moving parts; it is “solid state.” Flash memory is also nonvolatile—it retains data even when the power is turned off.
- Flash memory media are available in three forms:
 - Some tablets, laptops, desktops, and servers feature a **solid-state drive (SSD)**, which uses flash memory to store data, instructions, and information.
 - **Flash memory cards**, or flash RAM cards, are removable and reusable storage media that are inserted into a flash memory slot in a digital camera, notebook, smartphone, or other mobile device.
 - A **USB flash drive** consists of a finger-size module of reusable flash memory that plugs into the USB ports of nearly any microcomputer.





Smart Card: pocket-size card with integrated circuits.

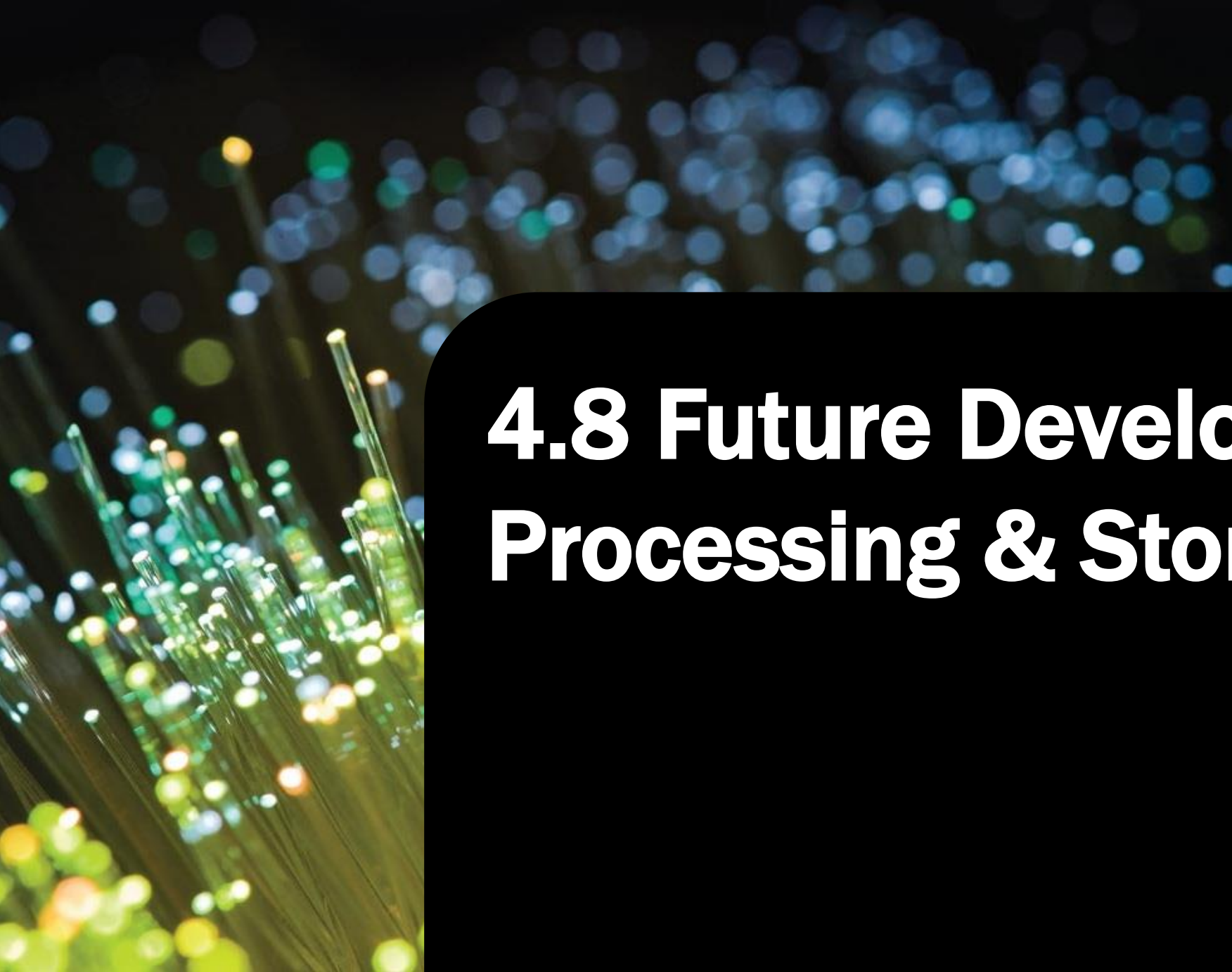
- Resembles a credit card but contains a microprocessor and memory chips
 - May function on three levels: credit, debit, and/or personal information
 - Storage capacity: around 10 MBs
 - Contact smart cards
 - Must be swiped through card readers
 - Can wear out from use
 - Contactless smart cards
 - Read when held in front of a low-powered laser





- Online Secondary Storage (**Cloud Storage**)
 - Allows you to use the Internet to back up your data
 - Sign up with a vendor and receive access to software and applications that allow you to upload your data to that company's server





4.8 Future Developments in Processing & Storage



Nanotechnology

Optical computing

DNA computing

Quantum computing

Better batteries

Description of Processing Technology

- Tiny machines work at a molecular level to make nanocircuits
- Uses lasers and light, not electricity
- Uses strands of synthetic DNA to store data
- Based on quantum mechanics and stores information using particle states
- Wireless charging of batteries, longer-lasting batteries



Future Developments in Secondary Storage

- Higher-density disks
 - Perpendicular recording technology: stacking magnetic bits vertically on the surface of a platter (instead of horizontally, as usual)
- Molecular electronics– storage at the subatomic level

